Lesson 12 Teacher's Key: Hearing It Happen

GameMaker - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

No new GameMaker mechanism this week - audio_play_sound() was already fully established by Lesson 3, so this lesson is purely about recognizing which existing branches (from Lessons 4, 8, 11) are the correct trigger points for one brand-new sound, and making an existing one mandatory.

Level 1 - Full Worked Solution

- sfx_wave is the one genuinely new sound this week, placed exactly where Lesson 11's nested if already lives.
- sfx_gameover, optional since Lesson 8, becomes mandatory starting today - if a student skipped it, this is the lesson where it has to go in.
- audio_play_sound()'s three arguments (sound, priority, loop) stay identical to Lesson 3's original call - no new parameter usage needed.

> ⚠ Some students' projects may have already fixed the "Stalemate" bug from Lessons 10-11 (an optional Level 3 goal), which replaces obj_bullet's inline win/wave-advance if/else with a shared wave_cleared() function called from BOTH obj_bullet's hit path and obj_enemy's escape path. If a student's project already has wave_cleared(), the sfx_wave call belongs ONCE, inside wave_cleared()'s else branch - not duplicated inline in obj_bullet's Collision event the way the base Level 1 solution (see the Handout's code) shows it. Since this depends entirely on whether a student reached that optional Level 3 goal, check each project's actual structure before telling a student where to add the call.

Why This Structure

- Because every trigger point (a collision, a wave transition, an escape) is already event-driven rather than polled, today's sound doesn't need any kind of replay-guard - it fires from code that only runs once per actual occurrence.
- Keeping every sound call physically adjacent to the game-state change it represents (wave advance, then immediately the matching sound) keeps this lesson's addition easy to audit later - a direct continuation of the Structure principle from Lesson 7's original bugfix.

Level 2 / Level 3 Talking Points

- Ask students to find all five audio_play_sound() calls now spread across obj_bullet and obj_enemy (sfx_hit, sfx_win, sfx_wave, sfx_gameover, sfx_lifelost) and name, for each, exactly which player action or game event causes it to run.
- For advanced students: ask what would need to change to give each enemy type (if the project ever added more than one) its own distinct hit sound - not required this week, a natural preview of expanding the Sound resource set later in the year.

Common Student Mistakes

- Placing the sfx_wave call outside the else branch, causing it to also fire on the true win.
- Forgetting the obj_gamemanager. prefix somewhere in the revised blocks, since this code still runs from obj_bullet and obj_enemy.
- Skipping sfx_gameover because it was optional before - it's mandatory as of this lesson.